# Worksheet 8: Graph neural networks part 2

## Numerical exercise: invariance versus equivariance

Consider three weather stations located in Athlone (A), Belfast (B), and Cork (C). We define the initial temperature vector as follows:

$$\mathbf{x} = \begin{bmatrix} A : 15 \\ B : 18 \\ C : 12 \end{bmatrix}$$

We now define a permutation operation $P$ that reorders the stations to $[B, A, C]$. The permuted input becomes:

$$P\mathbf{x} = \begin{bmatrix} B : 18 \\ A : 15 \\ C : 12 \end{bmatrix}$$

### 1. Permutation invariance (global diagnostics)

Let the function $f(\mathbf{x})$ represent the maximum temperature recorded across all stations.

- Applying the function to the original order: $f(\mathbf{x}) = \max(15, 18, 12) = 18$
- Applying the function to the permuted order: $f(P\mathbf{x}) = \max(18, 15, 12) = 18$

Because $f(P\mathbf{x}) = f(\mathbf{x})$, the output is invariant. This is useful for global metrics like total atmospheric energy.

### 2. Permutation equivariance (local forecasting)

Let the function $g(\mathbf{x})$ represent a Graph Neural Network layer that updates each station temperature. In this toy example, the update for a station is the average of its own value and its neighbors. Assume the output values are:

$$g(\mathbf{x}) = \begin{bmatrix} A_{new} : 16.5 \\ B_{new} : 13.5 \\ C_{new} : 15.0 \end{bmatrix}$$

If we permute the input, the output must change in a predictable way:

$$g(P\mathbf{x}) = \begin{bmatrix} B_{new} : 13.5 \\ A_{new} : 16.5 \\ C_{new} : 15.0 \end{bmatrix}$$

Because $g(P\mathbf{x}) = Pg(\mathbf{x})$, the function is equivariant. The forecast for Athlone remains 16.5 regardless of whether Athlone is the first or second element in the data matrix.

## Slide 10: GCN

Aim to compute:

$$h_u = \phi\left(\mathbf{x}_u, \bigoplus_{v \in N_u} c_{vu}\psi(\mathbf{x}_v)\right) \tag{1}$$

This is the embedding value for node $u$.

We represent our cities with two features: $[Temp, Precip]$. Let $x_A = [10, 0]$, $x_B = [8, 2]$, and $x_C = [12, 1]$.

### 1. Define $\psi$

We define the message function $\psi(x_v) = W_\psi x_v + b_\psi$. For simplicity, let $W_\psi = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (identity) and $b_\psi = \begin{bmatrix} 0.5 \\ 0.1 \end{bmatrix}$.

Calculating message from Belfast ($B$):

$$\psi(x_B) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ 2 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 8.5 \\ 2.1 \end{bmatrix}$$

### 2. Aggregation ($\bigoplus$)

We define $\bigoplus$ as $\sum$ and set constants $c_{BA} = 0.5, c_{CA} = 0.5$. Assume $\psi(x_C) = [12.5, 1.1]$. The total value for Athlone (sometimes called the message $m_A$) is:

$$m_A = 0.5 \begin{bmatrix} 8.5 \\ 2.1 \end{bmatrix} + 0.5 \begin{bmatrix} 12.5 \\ 1.1 \end{bmatrix} = \begin{bmatrix} 10.5 \\ 1.6 \end{bmatrix}$$

### 3. The Update Step ($\phi$)

To produce the final forecast $h_A$, we concatenate the local features $\mathbf{x}_A$ and the aggregated message $\mathbf{m}_A$, then apply a learnable update layer:

$$h_A = \mathbf{W}_\phi \begin{bmatrix} \mathbf{x}_A \\ \mathbf{m}_A \end{bmatrix} + \mathbf{b}_\phi$$

Let the update weights and bias be:

$$\mathbf{W}_\phi = \begin{bmatrix} 0.5 & 0.0 & 0.5 & 0.0 \\ 0.0 & 1.0 & 0.0 & 1.0 \end{bmatrix}, \quad \mathbf{b}_\phi = \begin{bmatrix} 2.0 \\ 0.0 \end{bmatrix}$$

$$h_A = \begin{bmatrix} 0.5(10.0) + 0.5(10.5) + 2.0 \\ 1.0(0.0) + 1.0(1.6) + 0.0 \end{bmatrix} = \begin{bmatrix} 12.25 \\ 1.6 \end{bmatrix}$$

Thus the 6-hour forecast for Athlone is 12.25°C with 1.6 mm of precipitation.

## Slide 12: full message passing with edge features

In this advanced example, we follow the formula:

$$h_u = \phi\left(\mathbf{x}_u, \bigoplus_{v \in N_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{uv})\right)$$

### 1. Node and edge features

We represent our cities with $[Temp, Precip]$ and edges with $[Dist, Angle]$.

- Athlone ($u$): $\mathbf{x}_u = [10, 0]$
- Belfast ($v$): $\mathbf{x}_v = [8, 2]$
- Edge ($e_{vu}$): $\mathbf{e}_{vu} = [140, 45]$ (140 km at 45°)

## 2. The message function ($\psi$)

We define $\psi$ as a linear transformation of the concatenated features:

$$\text{Triplet} = [\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{vu}] = [10, 0, 8, 2, 140, 45]$$

Let $\mathbf{W}_{msg}$ be a $2\times6$ matrix and $\mathbf{b}_{msg} = [0, 0]$. To keep calculations simple, we use a "weight mask" that focuses on the temperature difference and distance:

$$\mathbf{W}_{msg} = \begin{bmatrix} -0.1 & 0 & 0.1 & 0 & -0.01 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & 0 \end{bmatrix}$$

The message from Belfast to Athlone is:

$$\psi_{v\to u} = \begin{bmatrix} (-0.1 \times 10) + (0.1 \times 8) + (-0.01 \times 140) \\ (0.5 \times 2) \end{bmatrix} = \begin{bmatrix} -1 + 0.8 - 1.4 \\ 1.0 \end{bmatrix} = \begin{bmatrix} -1.6 \\ 1.0 \end{bmatrix}$$

## 3. Aggregation ($\bigoplus$)

Assuming Cork ($w$) sends a message $\psi_{w\to u} = [0.6, 0.4]$, the total message for Athlone is:

$$\mathbf{m}_u = \sum \psi = \begin{bmatrix} -1.6 \\ 1.0 \end{bmatrix} + \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} = \begin{bmatrix} -1.0 \\ 1.4 \end{bmatrix}$$

## 4. The update function ($\phi$)

Finally, we update Athlone's state using a weighted sum of its current state and the message, plus a bias:

$$h_u = \mathbf{W}_{upd}[\mathbf{x}_u, \mathbf{m}_u] + \mathbf{b}_{upd}$$

Using $\mathbf{W}_{upd} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$ and $\mathbf{b}_{upd} = [0, 0.5]$:

$$h_u = \begin{bmatrix} 10 - 1.0 \\ 0 + 1.4 \end{bmatrix} + \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 9.0 \\ 1.9 \end{bmatrix}$$

The forecast for Athlone is 9.0°C and 1.9 mm precipitation.